Finite Mathematics 2
Short Guide to Conditional Probability and Bayes' Rule

RECALL	DEFINITION
Sample Space	The set of all possible outcomes under consideration
Event	A specific subset of the sample space which produces a defined result
Probability	The likelihood of an event occurring within the sample space. This is found by: $\text{Probability} = \frac{\text{Number of Desired Outcomes}}{\text{Number of Possible Outcomes}}$ or, formally as $P(E) = \frac{n(E)}{n(S)}$
Mutually Exclusive Events	Events are called mutually exclusive if they cannot occur simultaneously. In other terms, if one event happens, all others cannot.

Concept: The Law of Total Probability
If we consider some set of mutually exclusive events E_1, E_2, …, E_n which, all-in-all form the entire sample space *S (see the figure below the next section),* then we say that:

$$\bigcup_{i=1}^{n}(E_i) = E_1 \cup E_2 \cup \ldots \cup E_n = S,$$

and by the equation for the cardinality of unions of mutually exclusive events, we say that

$$\sum_{i=1}^{n} n(E_i) = n(E_1) + n(E_2) + \ldots + n(E_n) = n(S)$$

and finally, dividing by $n(S)$, we arrive at:

Law of Total Probability

$$\sum_{i=1}^{n} P(E_i) = P(E_1) + P(E_2) + \ldots + P(E_n) = P(S) = 1$$

Formally, this can be stated simply as:
For some set of events E_1, E_2, … E_n whose union forms the sample space *S*, then the total probability of these events is 1.

Additional: Extension of the Law of Total Probability
Any other subset *F* of the sample space *S* beyond E_1 through E_n can be formed entirely from the union of its intersections with E_1 to E_n. Consider the figure below. The event *A* is formed entirely by its intersections with events B through B_n. Formally, we say:

Law of Total Probability for a Certain Event

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots + P(A \cap B_n) = \sum_{i=1}^{n} P(A \cap B_i)$$

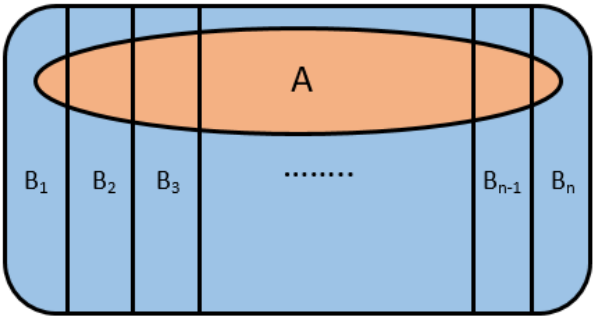


Figure: The Law of Total Probability
Notice that the union of events B_1 through B_n is the entire sample space.

Concept: Conditional Probability
Events are called *conditional* if the occurrence of one event directly affects the probability of the occurrence of others. For instance, the occurrence of daylight directly affects the occurrence that a plant grows by 1 cm.

When we consider conditional events, we do not look at the entire sample space. Instead, we look at the condition as a "relative sample space" and the intersection of this condition with the desired event as the new "relative event". That is,

Definition of Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

where B is the condition and A is the desired event

Such is read as "*The probability of A given B is the probability of their intersection over the probability of B.*" Another way of reading the left hand side is "*Given B, the probability of A*".

From this, a new form of the Law of Total Probability can be formulated. Multiplying both sides by P(B), we find that:

$$P(A \mid B) \cdot P(B) = P(A \cap B)$$

Recall that we also found:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \ldots + P(A \cap B_n) = \sum_{i=1}^{n} P(A \cap B_i)$$

Thus, substituting our newly found formula into each intersection probability, we find:

Alternative Form of Total Probability

$$P(A) = \sum_{i=1}^{n} [P(A \mid B_i) \cdot P(B_i)]$$

Concept: Bayes' Theorem
States that the probability of an event A given a condition B can be found using the probability of B using A as the condition. Consider the definition of conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

And consider A as the condition instead of B. Then, we get

$$P(B \mid A) = \frac{P(B \cap A)}{P(A)}$$

Multiplying both sides by P(A),

$$P(B \mid A) \cdot P(A) = P(B \cap A)$$

And since $B \cap A = A \cap B$,

$$P(B \mid A) \cdot P(A) = P(A \cap B)$$

Substituting that into P(A | B), we find the formal statement for Bayes' Theorem for two events:

Bayes Theorem for Two Events:

$$P(A \mid B) = \frac{P(A) \cdot P(B \mid A)}{P(B)}$$

If we set A to be a specific event among multiple, we can represent it instead with A_i then we can represent this as:

$$P(A_i \mid B) = \frac{P(A_i) \cdot P(B \mid A_i)}{P(B)}$$

If we apply the formula found earlier as $P(A) = \sum_{i=1}^{n} [P(A \mid B_i) \cdot P(B_i)]$, and we switch around A and B to find $P(B) = \sum_{i=1}^{n} [P(B \mid A_i) \cdot P(A_i)]$ we may find that the generalized form of Bayes' Theorem is:

Generalized Bayes' Theorem

$$P(A_i \mid B) = \frac{P(A_i) \cdot P(B \mid A_i)}{\sum_{i=1}^{n} (P(B \mid A_i) \cdot P(A_i)}$$

Reference Examples:

1. Suppose a sample space *S* is partitioned into two mutually exclusive events, B_1 and B_2. A subset event A overlaps with them such that the probability of its intersection with B_1 is 0.15 and with B_2 is 0.35. What is the total probability of event A?

Solution:

From the given, we find that $P(A \cap B_1) = 0.15$ and $P(A \cap B_2) = 0.35$. From the law of total probability, since the events B_1 and B_2 are defined to partition the entire sample space, we can invoke $P(A) = \sum_{i=1}^{n} P(A \cap B_i)$ and state that:

$$P(A) = P(A \cap B_1) + P(A \cap B_2) = 0.15 + 0.35 = \mathbf{0.50}.$$

2. An electronics manufacturer sources circuit boards from two suppliers: Supplier 1 (B_1) and Supplier 2 (B_2). Supplier 1 provides 70% of the boards and Supplier 2 provides the remaining 30%. The defect rate for Supplier 1 is 4%, and for Supplier 2 it is 9%. Calculate the total probability that a randomly chosen board is defective.

Solution:

Let A be the event of a product being defective, B_1 be the event of a product being from Supplier 1, and B_2 be the event of a product being from Supplier 2. From the given, we know that $P(B_1) = 0.70$ and

$P(B_2) = 0.30$. Also, we know that 4% of products from Supplier 1 are defective. This can likewise be stated as "Given that a product is from Supplier 1, there is a 4% chance it is defective." This directly translates to $P(A \mid B_1) = 0.04$ *(see discussion for understanding translation)* and, similarly, we see from the given that $P(A \mid B_2) = 0.09$. The question then asks for $P(A)$, and we know from our discussion that we can find this by $P(A) = \sum_{i=1}^{n} [P(A \mid B_i) \cdot P(B_i)]$. Solving, we find:

$$P(A) = P(A \mid B_1) \cdot P(B_1) + P(A \mid B_2) \cdot P(B_2) = 0.70 \cdot 0.04 + 0.30 \cdot 0.09 = \mathbf{0.55}$$

3. A university course has students split across three lecture sections: Section 1 holds 30%, Section 2 holds 50%, and Section 3 holds 20% of the students. The historical pass rates for each section are 0.80, 0.70, and 0.90 respectively. What is the overall pass rate for the entire course cohort?

Solution:
Similarly to the last example, we can automatically assign events. If A is the event of passing, and B_n is the event of a student being in Section n, then we find that:
$P(B_1) = 0.30, P(B_2) = 0.50, P(B_3) = 0.20, P(A \mid B_1) = 0.80, P(A \mid B_2) = 0.70,$ and $P(A \mid B_3) = 0.90$
The question asks for the probability of passing, so we apply the formula $P(A) = \sum_{i=1}^{n} [P(A \mid B_i) \cdot P(B_i)]$ and find that:

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + P(A \mid B_3)P(B_3) = 0.3 \cdot 0.8 + 0.5 \cdot 0.7 + 0.2 \cdot 0.9 = \mathbf{0.77}$$

4. A medical condition affects 1% of a given population. A diagnostic test yields a positive result for an affected person with a probability of 0.95. If the overall probability of any given person testing positive in the population is 0.0394, determine the probability that a person actually has the condition given that they tested positive.

Solution:
From the given, letting A denote the event of having the condition and B denote the event of testing positive. From the given, $P(A) = 0.01$, $P(B \mid A) = 0.95$, and $P(B) = 0.0394$. The problem asks for $P(A \mid B)$, so we must use Bayes' Theorem for two events, which we have previously found and stated as $P(A \mid B) = \dfrac{P(A) \cdot P(B \mid A)}{P(B)}$. Substituting values, we find that:

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(B)}{P(A)} = \frac{0.95 \cdot 0.01}{0.0394} = \mathbf{0.2411}$$

Suggested Practice:
Three factories supply widgets to a market: Factory 1 supplies 20%, Factory 2 supplies 50%, and Factory 3 supplies 30%. Their respective defect rates are 1%, 3%, and 2%. If a randomly picked widget is found to be defective, what is:

- (a) the probability that it originated from Factory 1?
- (b) the probability that it is from either Factory 2 or 3?
- (c) the probability that a product not defective comes from Factory 3?